INVESTMENTS

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CHAPTER 14

Bond Prices and Yields

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · Bond Characteristics

A bond is the "IOU" of a borrowing arrangement: the issuer sells the bond to the lender for cash and is obligated to make specified payments on specified dates. A typical coupon bond pays semi-annual interest for the bond's life, then repays face (par) value at maturity.

Government of Canada Bonds

Treasury notes/bonds are direct federal obligations with essentially zero default risk, differing mainly by original maturity (notes: 2–10 years; bonds: 10+ years).

Corporate Bonds and Indentures

Corporate bonds are issued under a **bond indenture,** the contract specifying the issuer's obligations and bondholders' rights. Common indenture features include: **call provisions** (issuer may repurchase at a stated call price before maturity — attractive to issuers when rates fall), **convertible provisions** (holder may exchange the bond for a fixed number of common shares), **put provisions** (the reverse of a call — the holder may force early redemption), and **sinking funds** (the issuer retires part of the issue each year, reducing the risk of a large single repayment at final maturity).

International and Innovative Bonds

Foreign bonds (issued by a foreign borrower but denominated in the investor's home currency) and Eurobonds (denominated in a currency other than that of the country where they are issued) are common international variants. Innovative structures include inverse floaters (coupon falls as interest rates rise), asset-backed bonds (payments tied to revenue from a specific project or pool of assets), and catastrophe bonds (coupon or principal is reduced if a specified catastrophic event occurs, effectively letting bond investors sell disaster insurance for extra yield).

LO2 · Bond Pricing

A bond’s price is the present value of all its promised cash flows (coupons plus final face value), discounted at the prevailing market interest rate for bonds of comparable risk and maturity (see Formulas page). When the coupon rate equals the market interest rate, the bond prices at exactly par value.

The Inverse Price–Yield Relationship

Bond prices and market interest rates move in **opposite directions:** when rates rise, the present value of a bond’s fixed future payments falls, so its price falls; when rates fall, price rises. This is the single most important relationship in fixed-income analysis, and it holds regardless of a bond’s specific coupon or maturity — though (as explored later in the course) the *magnitude* of the price response to a rate change varies systematically with maturity and coupon.

Premium and Discount Bonds

- **Par bond —** coupon rate = market rate; price = face value.
- **Premium bond —** coupon rate > market rate; price > face value (investors pay extra for the above-market coupon).
- **Discount bond —** coupon rate < market rate; price < face value.

LO3 · Bond Yields

Yield to Maturity (YTM)

YTM is the single interest rate that, when used to discount a bond's remaining promised cash flows, produces a present value exactly equal to its current market price (see Formulas page). It is the standard measure of a bond's average rate of return **if held to maturity**, and it implicitly assumes any coupons received are reinvested at that same rate — an assumption that may not hold in practice if reinvestment rates change over the bond's life.

Current Yield

Current yield is simply annual coupon payment ÷ current bond price — a much simpler, but incomplete, measure that ignores any built-in capital gain or loss the bondholder will realize as the price converges to par value at maturity.

The Coupon Rate / Current Yield / YTM Ordering

For a **premium bond:** coupon rate > current yield > YTM (current yield is diluted by the high price paid; YTM further accounts for the built-in capital *loss* as price falls to par by maturity). For a **discount bond,** this ordering fully reverses: coupon rate < current yield < YTM.

Yield to Call

For a callable bond, yield to call replaces the maturity date and value with the call date and call price in the YTM calculation — the relevant return measure *if* the bond is called at the issuer's option. Yield to call is especially relevant for premium bonds trading above their call price, since the issuer has a strong incentive to call and refinance at a lower rate.

LO4 · Bond Prices Over Time

A bond purchased at a premium or discount will see its price gradually converge toward par value as maturity approaches (assuming market interest rates stay constant), **purely from the passage of time** — not from any change in market conditions. This predictable, priced-in convergence is distinct from price changes caused by subsequent shifts in market interest rates.

Realized Compound Return vs. Yield to Maturity

YTM equals a bond's *actual* realized return only if every coupon is reinvested at exactly that same YTM rate. If reinvestment rates differ (e.g., market rates change during the holding period), the investor's **realized compound return** will differ from the YTM calculated at purchase. This distinction matters most for long-maturity bonds, where far more of total return depends on reinvested coupon income.

After-Tax Returns

In Canada, both interest income and any capital gain on a bond are generally taxable, but at different rates (capital gains benefit from a 50% inclusion rate federally). An **original issue discount (OID) bond** — issued below par with the price appreciation over its life effectively substituting for some coupon income — has special tax rules requiring the built-in appreciation to be accrued and taxed annually as interest income, rather than deferred entirely until sale.

LO5 · Default Risk and Bond Pricing

Unlike most government bonds, corporate bond payments are **not certain** — they depend on the ongoing financial health of the issuer. This uncertainty is **default (credit) risk.**

Credit Rating Agencies

Moody’s, Standard & Poor’s, Dominion Bond Rating Service (DBRS), and Fitch all assign letter-grade credit ratings to corporate and government bond issuers. The top rating (AAA/Aaa) is awarded to only about a dozen firms globally. Bonds rated BBB/Baa or above are **investment-grade;** below that threshold, bonds are **speculative-grade (junk bonds).** Empirically, nearly **half of all bonds rated CCC at issue by S&P have defaulted within 10 years** — but even highly rated bonds are not risk-free: **WorldCom sold $11.8 billion of investment-grade bonds in 2001, then filed for bankruptcy just one year later, with bondholders losing more than 80% of their investment.**

Junk Bonds

Before 1977, virtually all junk bonds were "fallen angels" — previously investment-grade bonds subsequently downgraded. In 1977, firms began issuing **original-issue junk** directly, an innovation credited largely to Drexel Burnham Lambert’s trader Michael Milken, who had built a network of willing junk-bond investors. High-yield bonds became notorious in the 1980s as financing vehicles for leveraged buyouts and hostile takeovers.

Determinants of Bond Safety

Rating agencies assess creditworthiness using financial ratios (coverage ratios, leverage ratios, liquidity ratios, profitability, and cash-flow-to-debt ratios) alongside qualitative factors (industry outlook, competitive position, management quality, and specific indenture provisions like collateral, seniority, and sinking funds).

The Default Premium

Bonds with greater default risk must offer a higher promised yield to compensate investors — the **default premium** (or credit spread) is the gap between a corporate bond's yield and that of an equivalent-maturity, default-free government bond. This spread widens during economic downturns (when default risk generally rises) and narrows during expansions.

Credit Default Swaps and Collateralized Debt Obligations

A **credit default swap (CDS)** is, in effect, an insurance contract against a bond issuer's default: the protection buyer pays a periodic premium, and the protection seller pays out if a specified default event occurs. A **collateralized debt obligation (CDO)** pools many bonds or loans and issues tranches with differing seniority (as introduced in Chapter 1's discussion of the 2008 crisis) — both instruments are central to how credit risk is packaged, transferred, and (as 2008 showed) sometimes mismeasured across the modern financial system.

Chapter 14 — Big-Picture Takeaways

- A bond's price is the present value of its promised coupon and principal payments, discounted at the prevailing market rate for its risk and maturity class.
- Bond prices and market interest rates always move in opposite directions — the single most important relationship in fixed-income analysis.
- Coupon rate vs. current yield vs. YTM ordering reveals whether a bond trades at a premium (coupon > current yield > YTM) or a discount (reversed).
- YTM assumes coupons are reinvested at the YTM rate itself; realized compound return can differ if actual reinvestment rates change over the holding period.
- Credit ratings (Moody's, S&P, DBRS, Fitch) classify bonds as investment-grade or speculative-grade (junk); even investment-grade ratings do not eliminate default risk, as WorldCom's 2002 bankruptcy dramatically showed.
- Riskier bonds must offer a default premium (credit spread) over comparable government bonds; CDS and CDOs are the primary instruments for transferring and repackaging that credit risk.

Bond pricing and yield calculations are the most heavily and consistently tested content in this chapter.

F1 | Bond Price (Present Value of Cash Flows)

Price = Σ [Coupon / (1+r)^t] + [Face Value / (1+r)^T]
= Coupon × Annuity Factor(r, T) + Face Value × PV Factor(r, T)

r = market interest rate per period; T = number of periods to maturity. For semi-annual bonds, use the semi-annual coupon, the semi-annual rate, and the number of half-year periods.

F2 | Current Yield

Current Yield = Annual Coupon Payment / Current Bond Price

A simpler, incomplete return measure that ignores any built-in capital gain or loss as the bond's price converges to par by maturity.

F3 | Yield to Maturity (YTM) — Solve for r

Price = Σ [Coupon / (1+r)^t] + [Face Value / (1+r)^T]
Solve for r given the bond's current market Price

The single discount rate that equates the present value of all remaining promised payments to the bond's current price. Requires a financial calculator, spreadsheet, or iterative solution — no closed-form algebraic solution exists.

F4 | Coupon / Current Yield / YTM Ordering

Premium bond: Coupon Rate > Current Yield > YTM
Discount bond: Coupon Rate < Current Yield < YTM

A quick internal-consistency check: if your calculated values don't follow this ordering given the bond's premium/discount status, recheck your work.

F5 | Default (Credit) Premium

Default Premium = YTM(corporate bond) − YTM(comparable-maturity government bond)

Compensates investors for bearing the corporate issuer's default risk. Widens in economic downturns; narrows in expansions.

EXAMPLE 14.2 | Bond Pricing

Given: An 8% coupon, 30-year bond, $1,000 par value, paying 60 semi-annual coupons of $40 each.

If the market rate is 8% annually (4% per half-year):

```
Price = $40 × Annuity Factor(4%, 60) + $1,000 × PV Factor(4%, 60) =
$904.94 + $95.06 = $1,000.00
```

The coupon rate equals the market rate, so the bond prices exactly at par — as expected.

If the market rate rises to 10% annually (5% per half-year):

```
Price = $40 × Annuity Factor(5%, 60) + $1,000 × PV Factor(5%, 60) =
$757.17 + $53.54 = $810.71
```

The bond's price falls by $189.29 as the market rate rises — a direct demonstration of the inverse price–yield relationship.

EXAMPLE 14.4 | Yield to Maturity and Yield Comparisons

Given: The same 8% coupon, 30-year bond is now selling at $1,276.76.

Step 1 — Solve for YTM (per half-year), then annualize:

```
$1,276.76 = $40 × Annuity Factor(r,60) + $1,000 × PV Factor(r,60)
→   r = 3.00% per half-year
Annualized YTM (simple interest, bond-equivalent basis) = 2 × 3.00%
= 6.09%
```

Step 2 — Current yield:

```
Current Yield = $80 / $1,276.76 = 6.27%
```

Step 3 — Confirm the premium-bond ordering:

```
Coupon Rate (8.00%) > Current Yield (6.27%) > YTM (6.09%)   ✓
```

This bond trades at a premium (price $1,276.76 > par $1,000), and the ordering confirms it: the coupon rate is highest, current yield is diluted by the high price, and YTM is lowest because it also captures the built-in capital loss as the price falls back to par by maturity.

PRACTICE PROBLEM | Crowfoot Fixed Income Partners

Scenario

You are a fixed-income analyst at Crowfoot Fixed Income Partners in Calgary, evaluating a corporate bond issued by a mid-cap Alberta energy services firm: a 6% coupon, 20-year bond, $1,000 par value, paying semi-annual coupons, currently priced at $891.36. The bond is rated BB by DBRS.

Part A — Bond Pricing Fundamentals

(i) Is this bond trading at a premium or a discount to par value? Explain how you know.

(ii) If market interest rates for bonds of this risk and maturity were to fall, would you expect this bond's price to rise or fall? Explain using the inverse price–yield relationship.

(iii) Calculate the bond's current yield.

Part B — Yield to Maturity

(i) Set up (but do not necessarily solve by hand) the equation you would use to find this bond's yield to maturity, given its price, coupon, and maturity.

(ii) Without solving precisely, state whether you expect the bond's YTM to be higher or lower than its current yield, and explain why, referencing the coupon/current yield/YTM ordering.

Part C — Credit Risk

(i) Is a BB-rated bond investment-grade or speculative-grade? Explain using the rating thresholds from LO5.

(ii) Explain, referencing the WorldCom example, why even an investment-grade rating does not fully eliminate default risk for a bondholder.

(iii) Would you expect this BB-rated bond's yield to be higher or lower than an otherwise identical AAA-rated bond's yield? Name this yield gap.

Part D — Realized Return vs. YTM

Suppose Crowfoot buys this bond today at its calculated YTM, planning to hold it to maturity.

(i) Under what specific assumption about coupon reinvestment would the firm's actual realized compound return exactly equal the YTM calculated at purchase?

(ii) If market interest rates fall significantly one year after purchase and stay low for the rest of the bond's life, would you expect Crowfoot's realized compound return to end up above or below the original YTM? Explain.

Hints & Guidance

Part A tests LO2 and Formula F2 (mirror Example 14.4’s current yield step). Part B tests LO3 and Formulas F3–F4. Part C tests LO5 directly. Part D tests LO4 — remember realized compound return depends on the ACTUAL reinvestment rate for coupons received, not the original YTM.